Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
M.Sc Degree Examination – APRIL 2021

MT101S – ALGEBRA AND TRIGONOMETRY

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. If α, β, γ are roots of the equation $x^3 + px^2 + qx + r = 0$ find the equation whose roots are $\alpha^3, \beta^3, \gamma^3$.

2. Prove that $\frac{e-1}{e+1} = \frac{\frac{1}{2!}+\frac{1}{4!}+\frac{1}{6!}+..\infty}{\frac{1}{1!}+\frac{1}{3!}+\frac{1}{5!}+..\infty}$

3. If A and B are Hermitian show that AB + BA is Hermitian.

4. Find the number of integers less that 600 and prime to it.

5. If $\tan\frac{x}{2} = \tanh\frac{x}{2}$ show that $\cos x \cosh x = 1$.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Prove that the equation $x^4 + 3x - 1 = 0$ has two real and two imaginary roots.

7. Sum to infinity the series $1+\frac{2^4}{2!}+\frac{3^4}{3!}+\frac{4^4}{4!}+..\infty$

8. Find the rank of the matrix A = $\begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{bmatrix}$

9. Find the highest power of 2 in 1000 !

10. Separate into real and imaginary parts $\tan^{-1}(x+iy)$.

SECTION - C (5X10=50)

Answer **ALL** Questions

11. (a) Solve $6x^5 - x^4 - 43x^3 + 43x^2 + x - 6 = 0$

(or)

(b) Find the real root of the equation $x^3 + 6x - 2 = 0$ using Horner's method.

12. (a) Sum to infinity the series $\frac{1.3}{2.4.6.8}+\frac{1.3.5}{2.4.6.8.10}+\frac{1.3.5.7}{2.4.6.8.10.12}+\ldots\infty$

(or)

(b) Show that $\frac{1}{1.2.3}+\frac{1}{3.4.5}+\frac{1}{5.6.7}+..\infty=\log 2-\frac{1}{2}$

13. (a) Find the value of a and b such that the equations
x + y + 2z = 2
2x – y + 3z = 2
5x – y + az = b have (i) no solution (ii) unique solution.

(or)

(b) Show that the matrix A = $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ satisfies Cayley-Hamilton theorem.

14. (a) Show that $n^{13} - n$ is divisible by 2730 if n is prime to 2730.

(or)

(b) Show that 10! – 32 is divisible by 143.

15. (a) Expand $\cos 6\theta$ interms of $\sin\theta$

(or)

(b) If $\sin(A+iB)=x+iy$ prove that

(i). $\frac{x^2}{\cosh^2 B}+\frac{y^2}{\sinh^2 B}=1$

(ii) $\frac{x^2}{\sin^2 A}-\frac{y^2}{\cos^2 A}=1$

****